

RESEARCH ARTICLE

Cumulative Adaptive Extended Kalman Filter for Robust UWB Positioning in Indoor Environments

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ABSTRACT

In complex indoor environments, the ultrawide band (UWB) positioning systems enhanced by Kalman filters (KFs) often struggle with large variations in accuracy and reliability due to outlier observations in the state system. This paper presents a cumulative adaptive extended KF (CA-EKF) to mitigate this issue by optimizing state estimation accuracy. The proposed technique adaptively adjusts process and measurement noise covariance matrices based on residual and innovation vectors, effectively suppressing filtering divergence and significantly enhancing positioning precision. Performance evaluation demonstrates the superiority of CA-EKF over traditional extended KF (EKF), innovation adaptive KF (IAKF), and Sage–Husa adaptive KF (SHAKF). Monte Carlo simulations under multiple outlier scenarios and nonlinear motion trajectories confirm CA-EKF's robustness, achieving lower positioning errors, smoother error evolution, and improved resilience against multiple simultaneous outliers. Quantitative results highlight that CA-EKF consistently outperforms other filters in terms of running-average root mean square error (RA-RMSE), mean squared error (MSE), and standard deviation (STD), reducing error accumulation over time and enhancing tracking accuracy. Notably, cumulative distribution function (CDF) analysis confirms that 99% of positioning errors with CA-EKF remain below 7.65 cm significantly outperforming competing methods. These results establish CA-EKF as a highly effective solution for high-precision UWB tracking applications in dynamic environments.

1 | Introduction

The positioning in indoor areas involves a diversity of sciences and technologies and requires accuracy that can range from a few meters to submillimeter precision. Unfortunately, classical Global Navigation Satellite Systems (GNSS) like GPS, GLONASS, and Galileo, fail to provide such precision levels, because of satellite signal deterioration caused by shadowing and multipath fading, incoherent propagation, dynamic environments, and severe attenuation imposed by indoor structural elements [1–6]. In medicine, for example, localization inside the human body requires precision ranging from a fraction of a millimeter for neurons in the brain to several centimeters in the gastrointestinal tract [1]. This issue triggered extensive research

efforts in indoor tracking and the localization of moving objects since the 1990s, which have expanded towards a wide range of innovative applications in the evolving smart world such as the Internet of Things (IoT), smart infrastructures, smart devices, transportation systems, robots, military applications, smart healthcare, and many more.

Different solutions have been proposed so far, such as Wi-Fi, Bluetooth Low Energy (BLE), ultrawide band (UWB), acoustic signals, and Radio Frequency Identification (RFID), with varying accuracy, power requirement, cost, and ease of deployment. References [7, 8] present a very interesting in-depth comparative study of these diverse indoor positioning system (IPS) technologies, highlighting the main challenges facing each of them.

UWB technology provides important advantages that distinguish it from other solutions, such as low system complexity, secure and robust communications, wide bandwidth, high wireless transmission rate, low transmission signal power, and superior positioning accuracy [9–12]. However, in dense and complex indoor environments, where transmitted signals are obstructed by moving people and a variety of objects, there are serious challenges that must be addressed to achieve the key features demanded by a UWB-based IPS, namely, faster data transfer and enhanced positioning accuracy. The most critical challenge arises from errors in non-line-of-sight (NLOS) situations, tracking signal interference in dense environments, and functioning in extreme conditions, where the systems may present a large RMSE in positioning and, thus, a serious deterioration in localization precision [1, 13, 14]. In addition, factors such as excess delay, clock drift, signal interference, and the computational time required for user position estimation represent significant challenges [15–17].

In [18], the authors provide a high-quality state-of-the-art review of recent advances in UWB positioning in recent years. They have clearly discussed the encountered challenges and the proposed solutions, including modern approaches that involve machine learning and sensor data fusion. Moreover, the same reference emphasized the need of standardizing the various proposed and adopted techniques, similar to GNSS systems.

The UWB system consists of anchor-tag communication through the UWB indoor channel and the calculation of distance values from each anchor using signal metrics such as time of arrival (TOA), angle of arrival (AOA), and time difference of arrival (TDOA), which are used for position estimation.

Yet, most UWB-based positioning systems use the concept of TOA ranging to determine the user position [18], which brings additional hardware and higher power consumption, thus limiting their cost-effectiveness relative to other techniques [19, 20]. Moreover, as the TOA technique exploits trilateration to determine the position of mobile users based on the range from the mobile unit to at least three (3D) anchors at known locations, it imposes the problem of accurate synchronization between the anchors and tags that should be resolved [18].

Various techniques can be used to estimate the position from the TOA observables. The one that is widely chosen in IPSs is filtering algorithms, which have been found to improve the positioning accuracy of the tag during movement, while generating another challenge related to the choice of filtering algorithm itself [21–23]. Research efforts made in this context showed that robust Kalman filtering algorithms provide better position estimates, thereby improving navigation positioning accuracy [24], and since then, many research efforts have been made to optimize their performance.

The limitation of Kalman filter (KF) in handling nonlinear systems [18] has led to the emergence of other improved KF-based versions, the most efficient of which are the extended KF (EKF) [25] and its enhanced type, the unscented KF (UKF) [26], that can evaluate the nonlinear state estimates by a linear approximation [27]. Subsequently, the particle filters (PFs), which represent a position estimation application of Bayesian filters, were used to address estimation problems encountered in the

EKF [27]. Nevertheless, PFs require significant computational sources and might not be appropriate for real-time applications where latency is crucial [28].

Additionally, robust Kalman filtering techniques such as the robust EKF (REKF) were proposed by Wang et al. [29], accompanied by hypothesis testing and robust estimation to provide relatively reliable optimized residuals and effective control of outlier influences caused by gross errors. Although this approach shows promise in addressing large errors, its computational complexity may make it difficult to implement in real time. Furthermore, the robust estimation parameters of the REKF must be carefully adjusted, which can be difficult in dynamic environments. In contrast, PFs use a set of particles to represent the state distribution, providing enhanced robustness in the presence of heavy-tailed noise distributions.

To further reduce the effect of outliers and increase the robustness of state estimation, a number of adaptive KF (AKF) techniques have also demonstrated their efficiency, such as the Bayesian, correlation, maximum likelihood (ML), and covariance matching (CM) methods [30–32]. Building upon these advancements, Poulou et al. [15] proposed the UWB-EKF position estimator and proved through intensive simulation results its effectiveness for multiple anchor UWB indoor localization compared to conventional UWB localization approaches. However, the average computation time for multiple anchors is increasing with respect to the number of anchors. With the aim of further improving UWB positioning accuracy and reliability in intricate indoor environments and diverse conditions, Rahul Ranjan et al. [33] proposed the integration of EKF with a low-pass filter (LPF) and proved its effectiveness and superiority to the standalone EKF. However, the evaluation's focus on a limited set of trajectory types may restrict the applicability of the results to more complex or diverse indoor environments. Continuing in this direction, Eang and Lee [34] proposed a Deep Neural Network-based EKF (DNN-EKF) approach for optimizing UWB localization performance. This method reduces distance loss significantly by leveraging DNNs to refine EKF predictions. However, as it requires prior training with environment-specific data, its generalizability in dynamic or unstructured environments remains limited. Nevertheless, the performance of the LPF-EKF combination can be sensitive to the filter settings, particularly in highly dynamic scenarios.

The present work adopts primarily the CM-based method that uses adaptive algorithms based on innovative estimates of the process (Q) and/or measurement (R) noise covariance matrices [35, 36]. During the last few years, the CM method has been applied to improve the performance of estimation techniques for indoor localization and object tracking in UWB localization. Xu et al. in [37] proposed a novel approach to calculate fading factors using innovation covariance. This method compensates for the impact of incorrect information by adjusting the error covariance using the fading factor. However, the study's focus on theoretical simulations, coupled the absence of real-world validation, limits its practical applicability.

The authors in [38] proposed a traditional innovative filtering algorithm to improve the filtering accuracy by an online update of the system noise variance matrix and measurement noise variance matrix. While the results are promising, the study does not go into

great detail about scalability or computational overhead in large-scale systems. Lv et al. in [39] and Liu et al. in [40] presented an efficient Sage-Husa adaptive Kalman filter (SHAKF) based on time-varying noise. It offers the possibility of measuring and correcting in real time the statistical characteristics of system noise and measurement noise, to reduce the error of the system model. The study would, however, benefit from a more thorough examination of the filter's resilience in various environmental circumstances. Recently, in [41], Sun et al. proposed an improved version of the innovation AKF (IAKF) algorithm to overcome the shortcomings of both classical attenuation filtering and innovation-based adaptive filtering. This technique detects anomalies using a chi-square test on the innovation and dynamically modifies the measurement noise. This particular approach, which is simpler and safer than conventional methods, reduces positioning errors by 7% while retaining its effectiveness in complex urban contexts, making it ideal for use in autonomous vehicles.

In light of these developments, we propose a novel UWB indoor positioning method referred to as the cumulative adaptive EKF (CA-EKF). This latter is mainly a CM-based technique using the adaptive estimation of both Q and R noise covariance matrices. The key idea of CA-EKF is that after the innovation vector-based chi-square test detects the abnormal observations, the noise covariance matrices Q and R are adaptively updated using the cumulative average of the residual and innovation vectors. This approach can remove the filtering divergence caused by outliers and significantly improve the positioning accuracy. In a positioning system using UWB probes, the performance of the proposed CA-EKF is proven using simulated data for various scenarios of outlier position and nonlinear motion trajectory. The results on simulated datasets confirm the stability and convergence of the CA-EKF. In addition, a comparative study shows that increased accuracy can be achieved by using the proposed technique compared to the IAKF, SHAKF, and traditional EKF techniques.

The remainder of the paper is presented as follows: The second part provides some background and outlines the challenges. The proposed CA-EKF algorithm is presented in Section 3, along with its modeling, derivation, and stability proof. Section 4 presents the numerical implementation and test results of CA-EKF for UWB positioning. Section 5 concludes the paper.

2 | EKF-Based UWB Positioning

The 2D distance measurement of UWB positioning can be expressed by [42, 43]

$$d_i = \sqrt{(x_i - x)^2 + (y_i - y)^2} + n_i, \quad (1)$$

where d_i , (x_i, y_i, z_i) , and (x, y, z) are, respectively, the distance measurements between the i th UWB sensor taken as an anchor (a_i) and one UWB sensor taken as a tag. n_i denotes the measurement noise given by

$$n_i \sim N(0, R), \quad (2)$$

where $N(0, R)$ designates the Gaussian distribution of the noise with zero mean and predetermined covariance R .

By referring to [44], the discrete-time nonlinear models of state and observation for UWB positioning EKF are given, respectively, by

$$X_k = f(X_{k-1}, U_k) + W_{k-1}, \quad (3)$$

and

$$Z_k = h(X_k) + N_k, \quad (4)$$

where $f(\cdot)$ and $h(\cdot)$ are the nonlinear state transition function and observation function related to distance measurement, respectively. X_k and X_{k-1} are the predicted and the estimated process state vectors, respectively, U_k is the control input vector, Z_k is the measurement vector, $N_k \sim N(0, R)$ is the observation noise, and $W_{k-1} \sim N(0, Q)$ is the white Gaussian noise process type with zero mean and predetermined covariance Q .

The EKF approaches discrete-time linear models by linearizing the state and observation models through the Jacobian matrices Φ_k and H_k of the partial derivatives with respect to X_k of the functions $f(\cdot)$ and $h(\cdot)$, respectively. Therefore, (3) and (4) are linearized, respectively, as follows [45, 46]:

$$X_k = \Phi_k X_{k-1} + W_{k-1}, \quad (5)$$

and

$$Z_k = H_k X_k + N_k, \quad (6)$$

where $\Phi_k = \frac{\partial f(X_{k-1}, U_k)}{\partial X_k}$ and $H_k = \frac{\partial h(X_k)}{\partial X_k}$ represent the state transition and observation matrices, respectively.

It should be noted that for the considered distance measurement vector (1) without measurement noise, H_k can be defined as [47, 48]

$$H_k = \begin{bmatrix} \frac{x_k - x_1}{d_{1,k}} & \frac{y_k - y_1}{d_{1,k}} & 0 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & 0 \\ \frac{x_k - x_n}{d_{n,k}} & \frac{y_k - y_n}{d_{n,k}} & 0 & 0 & \dots & 0 \end{bmatrix}, \quad (7)$$

where $d_{i,k}$ is the measured range distance between a tag and the i th anchor and n indicates the number of anchors.

The distance measurements are obtained from the UWB sensor positioning system using the TOA method between three anchors (a_1, a_2, a_3) and one tag for two-dimensional positioning. Considering the tag moving on a nonlinear trajectory and constant turn rate and velocity (CTRV) as the motion model [48], then the state vector is given by

$$X_k = [x_k \ y_k \ \theta_k]^T, \quad (8)$$

and the nonlinear state transition of discrete sampling period T is

$$f(X_{k-1}, U_k) = \begin{bmatrix} \frac{v}{\omega} (\sin(\omega T + \theta_{k-1}) - \sin(\theta_{k-1})) + x_{k-1} \\ -\frac{v}{\omega} (\cos(\omega T + \theta_{k-1}) - \cos(\theta_{k-1})) + y_{k-1} \\ \omega T + \theta_{k-1} \end{bmatrix}, \quad (9)$$

where (x_k, y_k) is the position of the target in the x - y plan, θ_k represents the orientation angle of the target, v is the constant velocity of the target, and ω is the constant turning rate.

The state transition matrices can be expressed as

$$\Phi_{k-1} = \begin{bmatrix} 1 & 0 & \frac{v(\cos(\omega T + \theta_{k-1}) - \cos(\theta_{k-1}))}{\omega} \\ 0 & 1 & \frac{v(\sin(\omega T + \theta_{k-1}) - \sin(\theta_{k-1}))}{\omega} \\ 0 & 0 & 1 \end{bmatrix}. \quad (10)$$

And H_k becomes a 3×3 matrix given by

$$H_k = \begin{bmatrix} \frac{x_k - x_1}{d_{1,k}} & \frac{y_k - y_1}{d_{1,k}} & 0 \\ \frac{x_k - x_2}{d_{2,k}} & \frac{y_k - y_2}{d_{2,k}} & 0 \\ \frac{x_k - x_3}{d_{3,k}} & \frac{y_k - y_3}{d_{3,k}} & 0 \end{bmatrix}. \quad (11)$$

The discrete EKF equations can be expressed as follows [45, 46].

2.1 | Prediction Step

Compute the a priori estimate of the state vector $\hat{X}_k^-$ and the a priori covariance matrix P_k^- :

$$\hat{X}_k^- = \Phi_{k-1} \hat{X}_{k-1}^+, \quad (12)$$

$$P_k^- = \Phi_{k-1} P_{k-1}^+ \Phi_{k-1}^T + Q_{k-1}. \quad (13)$$

2.2 | Kalman Gain Computation

Compute the Kalman gain K_k :

$$K_k = P_k^- H_k^T [H_k P_k^- H_k^T + R_k]^{-1}. \quad (14)$$

2.3 | Update Step

Compute the state estimate $\hat{X}_k^+$ and the posteriori covariance matrix P_k^+ :

$$\hat{X}_k^+ = \hat{X}_k^- + K_k (z_k - H_k \hat{X}_k^-), \quad (15)$$

$$P_k^+ = (I - K_k H_k) P_k^-. \quad (16)$$

3 | Cumulative Adaptive EKF

In most cases, the optimal state estimation of EKF is obtained based on specific parameters governed by dynamic and stochastic models. In particular, the choice of the noise covariances Q_k and R_k of process and measurement, respectively, is crucial because of their direct impact on the stability and convergence of the Kalman filtering process. Relative to this context, the adaptive approach CA-EK is proposed in this work to ensure that any unforeseen and outlier change in the stochastic properties of the

noise, previously defined by a particular criterion [41, 49], is followed by an instantaneous adjustment of the R_k and Q_k values, satisfying variable environments necessities. Consequently, the proposed CA-EKF technique is particularly well suited for applications like UWB-based indoor localization because it presents actually cumulative adaptation mechanism that adaptively updates R_k and Q_k by utilizing the cumulative mean of residual and innovation vectors, thus ensuring enhanced stability, robustness to outliers, and adaptability to dynamic noise.

Afterwards, it is worth to note that CA-EKF operates within the framework of a standard EKF, which is used to estimate the state of a nonlinear dynamic system that is described by the discrete-time state and observation models given by Equations (3) and (4), respectively.

3.1 | Cumulative Adaptation Process

The major innovation of CA-EKF is its cumulative adaptive process, which dynamically updates the noise covariance matrices Q_k and R_k , based on the cumulative average of the residual and innovation vectors. This process comprises the following steps:

- *Innovation vector:*

The innovation vector r_k^+ is defined as the difference between the actual measurement Z_k and the predicted measurement $H_j \hat{X}_j^+$:

$$r_k^+ = Z_k - H_k \hat{X}_k^+, \quad (17)$$

where H_k is the observation matrix and $\hat{X}_k^+$ is the a posteriori state estimate.

- *Residual vector:*

The residual vector r_k^- is defined as the difference between the actual measurement Z_k and the predicted measurement $H_j \hat{X}_j^-$:

$$r_k^- = Z_k - H_k \hat{X}_k^-, \quad (18)$$

where $\hat{X}_k^-$ is the a priori state estimate.

- *Cumulative mean of innovation and residual vectors:*

To adaptively update the noise covariance matrices, the CA-EKF computes the cumulative mean of the innovation vector and that of the residual vector over time. The cumulative mean of the innovation vector C_k^+ is given by

$$C_k^+ = \frac{1}{k} \sum_{i=1}^k \left[\left(\frac{1}{m} \sum_{i=1}^m |r_i^+| \right)_{m=1:k} \right], \quad (19)$$

where m is the number of measurements at each time step.

In a similar manner, the cumulative mean of the residual vector C_k^- is determined by

$$C_k^- = \text{diag} \left[\frac{1}{k} \sum_{i=1}^k \left[\left(\frac{1}{m} \sum_{i=1}^m |r_i^-| \right)_{m=1:k} \right] \right]. \quad (20)$$

- *Adaptive update of noise covariance matrices:*

The cumulative means of innovation and residual vectors are used to update the noise covariance matrices Q_k and R_k .

The updated R_k for measurement noise is given by

$$R_k = (1 - d_k) \text{diag}(C_k^+) + d_k R_{k-1}, \quad (21)$$

where d_k is a weighting factor that regulates how much the cumulative mean influences the update that is given by

$$d_k = \frac{(1 - b)}{1 - b^{k+1}}, \quad (22)$$

where b is a factor of forgetting ($0 < b < 1$).

The process covariance matrix Q_k in the KF can be adjusted using the predicted measurement residuals and the Kalman gain matrix, based on the cumulative mean. The updated Q_k is given by

$$Q_k = (1 - d_k) \left(K_k [C_k^-]^T C_k^- K_k^T \right) + d_k Q_{k-1}. \quad (23)$$

- *Outlier detection and threshold selection:*

The adopted criterion for detecting aberrant variations of the EKF for UWB positioning is the Mahalanobis distance (MD) of the measured innovation vector defined as [50, 51]

$$MD^2(r_k^+) = [r_k^+]^T [H_k P_k^- H_k^T + R_k]^{-1} r_k^+. \quad (24)$$

When the criterion $MD^2(r_k^+)$ satisfies the higher inequality $MD^2(r_k^+) > \chi_\alpha^2(n)$, then the observed distance Z_k is considered an outlier. Otherwise, $MD^2(r_k^+) \leq \chi_\alpha^2(n)$, Z_k is the normal value, where $\chi_\alpha^2(n)$ is the statistical detection threshold of a confidence probability of the MD at the significance level α of the chi-square test with n -degree of freedom (n -DOF) [50, 51]. In UWB positioning, Z_k is a scalar and therefore follows a χ_α^2 distribution with 1-DOF. The probability of a random event is given as

$$Pr(MD^2(r_k^+) > \chi_\alpha^2(n)) = \alpha. \quad (25)$$

The outlier detection threshold is chosen as a function of the desired confidence level α and the degrees of freedom n , which correspond to the dimension of the measurement vector. For example, with $n = 1$, for a confidence level of 95% ($\alpha = 0.05$) and 99% ($\alpha = 0.01$), the chi-square threshold $\chi_\alpha^2(n)$ is, respectively, around 3.84 and 6.63. In general, the choice of α depends on the specific application and the acceptable false alarm rate (i.e., falsely reporting measurements as outliers). A lower α translates into a higher threshold, which reduces the probability of false alarm but risks some outliers going unnoticed. Conversely, a higher α results in a lower threshold, increasing sensitivity to outliers and the risk of false alarm [48, 51].

Once the outlier observation is detected, the two covariance matrices R_k and Q_k can be adjusted using the proposed approach based on the cumulative mean of the innovation and residual vectors, respectively.

- *Implementation of the proposed CA-EKF:*

The implementation of the proposed CA-EKF method is summarized in Algorithm 1.

4 | Numerical Implementation of CA-EKF for UWB Positioning

This section presents the numerical implementation of the proposed CA-EKF algorithm for UWB-based indoor localization. The simulation parameters are summarized in Table 1.

4.1 | Target Motion and Trajectory Scenarios

Figure 1 illustrates the 2D trajectory of the target along with the three anchor positions.

The target follows a CTRV model under two different trajectory scenarios:

- Traj1 (left of Figure 1): The target follows a smooth, nonlinear trajectory without maneuvers over the entire simulation.
- Traj2 (Right of Figure 1): The target undergoes two maneuvering phases:
 - 0–60s: Straight and uniform motion.
 - 61–180s: Turning maneuver, modifying its trajectory.
 - 181–260s: Resumption of uniform motion before executing a final turn.

The anchor positions are strategically selected to maintain consistent localization conditions across both trajectories, facilitating an in-depth evaluation of UWB positioning accuracy under maneuvering and nonmaneuvering conditions.

4.2 | Handling Outliers in Simulation Scenarios

To assess the accuracy of the proposed CA-EKF technique in handling finite-duration outlier observations, four target tracking simulation scenarios are considered. The latter ones are defined based on the relative positions of outliers with respect to each anchor (a_i) as follows:

- A. *Simultaneous outliers affecting all anchors* (T_{a_1, a_2, a_3}):

The abnormal information from UWB positioning is attributed to simultaneous outliers with finite-duration affects (T_{a_1, a_2, a_3}). Here, outlier observations persist for 60s within the common period of 90–150s across all three anchors a_i .

- B. *Pairwise simultaneous outliers* ($T_{a_1, a_2}, T_{a_1, a_3}, T_{a_2, a_3}$):

In this scenario, two anchors are affected by outliers at the same time, with each event lasting for 30s:

- T_{a_1, a_2} occurs between 60 and 90s (affecting a_1 and a_2).
- T_{a_1, a_3} occurs between 100 and 130s (affecting a_1 and a_3).

Algorithm 1: Proposed CA-EKF method implementation

Input: $X_0, Q_0, R_0, P_0, b = 0.2$
One-step $k = 1$
 $P_{k-1}^+ = P_0; \hat{X}_{k-1}^+ = X_0; Q_k = Q_0; R_k = R_0$
Output: $\hat{X}^+, P^+$
begin
 for $k = 2:N$ **do**
 $\hat{X}_k^- = \Phi_{k-1} \hat{X}_{k-1}^+$
 $P_k^- = \Phi_{k-1} P_{k-1}^+ \Phi_{k-1}^T + Q_k$
 $r_k^- = Z_k - H_k \hat{X}_k^-$
 $K_k = P_k^- H_k^T [H_k P_k^- H_k^T + R_k]^{-1}$
 $\hat{X}_k^+ = \hat{X}_k^- + K_k (Z_k - H_k \hat{X}_k^-)$
 $P_k^+ = (I - K_k H_k) P_k^-$
 $r_k^+ = Z_k - H_k \hat{X}_k^+$
 if $MD^2(r_k^+) > \chi_\alpha^2(1)$ **then**
 $d_k = (1 - b)/(1 - b^{k+1})$
 $C_k^- = \text{diag} \left[\frac{1}{k} \sum_{i=1}^k \left[\left(\frac{1}{m} \sum_{i=1}^m |r_i^-| \right)_{m=1:k} \right] \right]$
 $C_k^+ = \frac{1}{k} \sum_{i=1}^k \left[\left(\frac{1}{m} \sum_{i=1}^m |r_i^+| \right)_{m=1:k} \right]$
 $Q_k = (1 - d_k)(K_k [C_k^-]^T C_k^- K_k^T) + d_k Q_{k-1}$
 $R_k = (1 - d_k) \text{diag}(C_k^+) + d_k R_{k-1}$
 else
 $Q_k = Q_0$
 $R_k = R_0$
 end if
 end for
end

TABLE 1 | Monte Carlo simulation parameters.

Parameter	Value
Sampling period	0.1 s
Simulation time	260 s
Number of Monte Carlo simulations	100
Anchor positions	$[a_1, a_2, a_3] = [(0, 0), (30, 0), (30, 30)]$
Initial state	$X_0 = [5 \text{ m}, 5 \text{ m}, 0 \text{ rad}]^T$
Velocity	$v = 0.2 \text{ m/s (Traj1)}, v = 0.16 \text{ m/s (Traj2)}$
Turning rate	$\omega = 0.02 \text{ rad/s (Traj1)}, \omega = 0.0185 \text{ rad/s (Traj2)}$
Initial process noise covariance matrix	$Q_0 = \text{diag} \left([\sigma_p^2, \sigma_p^2, \sigma_p^2] \right), \sigma_p = 10^{-4} \text{ m}$
Initial measurement noise covariance matrix	$R_0 = \text{diag} \left([\sigma_m^2, \sigma_m^2, \sigma_m^2] \right), \sigma_m = 0.08 \text{ m}$
Noise amplification factor	$\lambda = 5$
Initial error covariance matrix	$P_0 = \text{diag}([10^{-3}, 10^{-3}, 10^{-3}])$

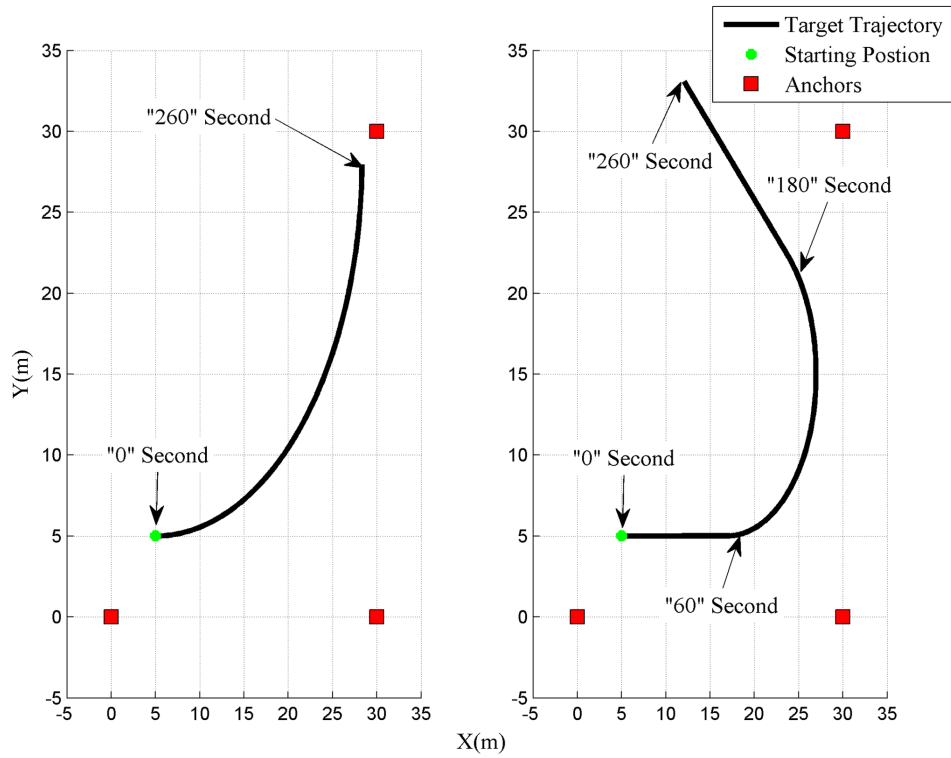


FIGURE 1 | 2D trajectory target in nonlinear motion: Left: Traj1 (without maneuvers)—The target follows a continuous curved path. Right: Traj2 (with maneuvers)—The target undergoes two maneuvering turns, altering its direction at specific time intervals.

- T_{a_1, a_2} occurs between 140 and 170 s (affecting a_2 and a_3).

C. *Isolated outlier scenario* ($T_{a_1}, T_{a_2}, T_{a_3}$):

In this case, three isolated outlier events occur at different time-stamps, each lasting for 30 s:

- T_{a_1} occurs between 60 and 90 s (affecting a_1).
- T_{a_2} occurs between 100 and 130 s (affecting a_2).
- T_{a_3} occurs between 140 and 170 s (affecting a_3).

D. *Combined scenario*:

The final scenario considers all possible outlier combinations. Specifically, outliers occur in all the cases examined in the previous three scenarios:

- Three isolated outlier periods: $(T_{a_1}, T_{a_2}, T_{a_3}) = (60-69, 70-79, 80-89)$ s.
- One simultaneous outlier period affecting all anchors: $(T_{a_1, a_2, a_3}) = (110-119)$ s.
- Three pairwise simultaneous outlier periods: $(T_{a_1, a_2}, T_{a_1, a_3}, T_{a_2, a_3}) = (90-99, 100-109, 120-129)$ s.

By selecting these finite-duration intervals, all possible outlier scenarios among the three anchors are analyzed.

In these four simulations, the effect of outlier observations is modeled by increasing the measurement noise amplitude within

the finite-duration interval T_X , as described by Equation (2), that is,

$$n_i \sim \lambda \times N(0, R), \quad (26)$$

where λ is the noise amplification factor.

4.3 | Initial Positioning Errors Before Applying CA-EKF

Figures 2 and 3 illustrate simulated positioning errors for two trajectories, Traj1 and Traj2, under different outlier scenarios: Simultaneous (three anchors), Simultaneous (two anchors), Isolated, and Combined.

While both figures follow similar trends, Traj1 (Figure 2) exhibits more pronounced error peaks in the Simultaneous (three anchors) and Combined cases, whereas in Traj2 (Figure 3), errors are more dispersed, particularly in the Simultaneous (two anchors) scenario. These differences highlight the trajectory-dependent impact of outliers on localization accuracy.

The results demonstrate the necessity of an effective outlier correction method, which will be applied next to evaluate its performance.

4.4 | Simulation Results

To assess the robustness of the proposed CA-EKF technique under outlier conditions, its performance is compared with

three alternative filtering methods: IAKF, SHAKF, and the conventional EKF. The evaluation is conducted using Monte Carlo simulations and relies on the following key performance metrics:

- *Positioning error versus time:* This metric examines the evolution of positioning errors under different outlier conditions, providing insights into both instantaneous and long-term tracking accuracy.

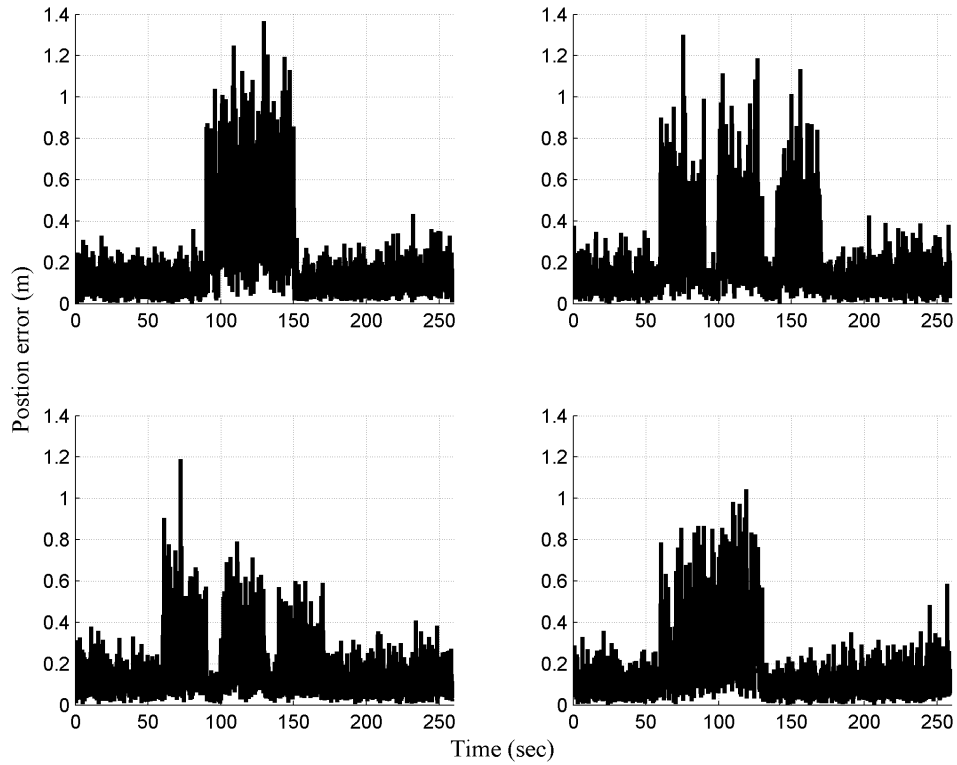


FIGURE 2 | UWB positioning error over time for different outlier scenarios with Traj1 (nonmaneuvering trajectory).

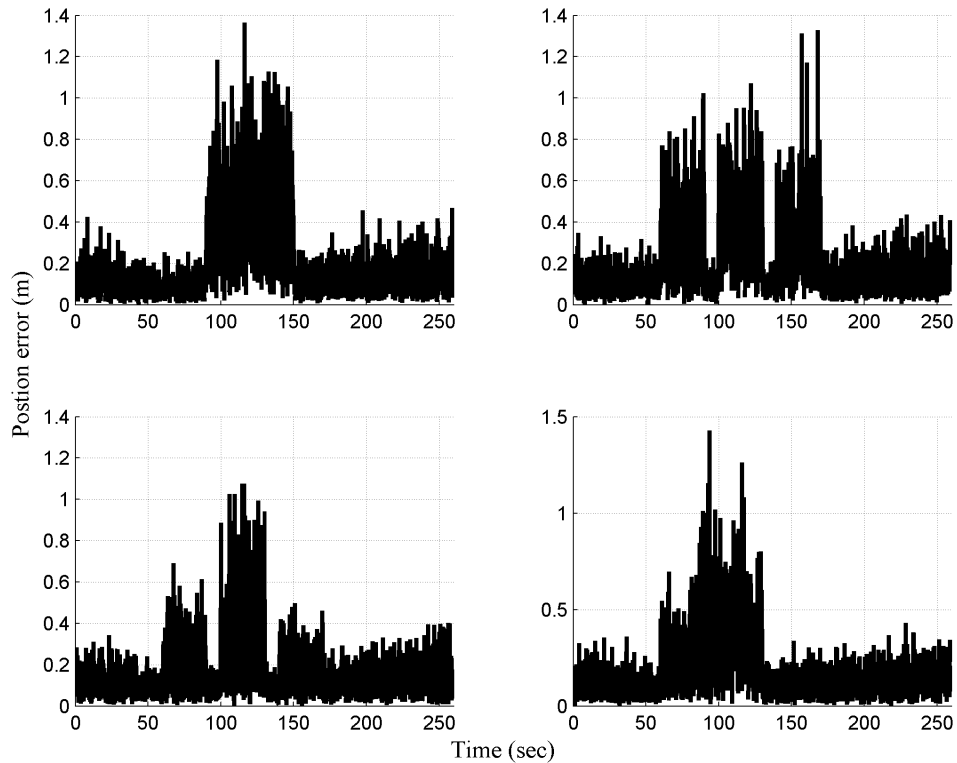


FIGURE 3 | UWB positioning error over time for different outlier scenarios with Traj2 (maneuvering trajectory).

- *RA-RMSE versus time*: The RA-RMSE provides a smoothed representation of tracking performance over time, offering a cumulative assessment of error trends.
- *CDF versus positioning error*: The CDF illustrates the probability distribution of positioning errors, helping to assess the reliability and consistency of the estimation methods.

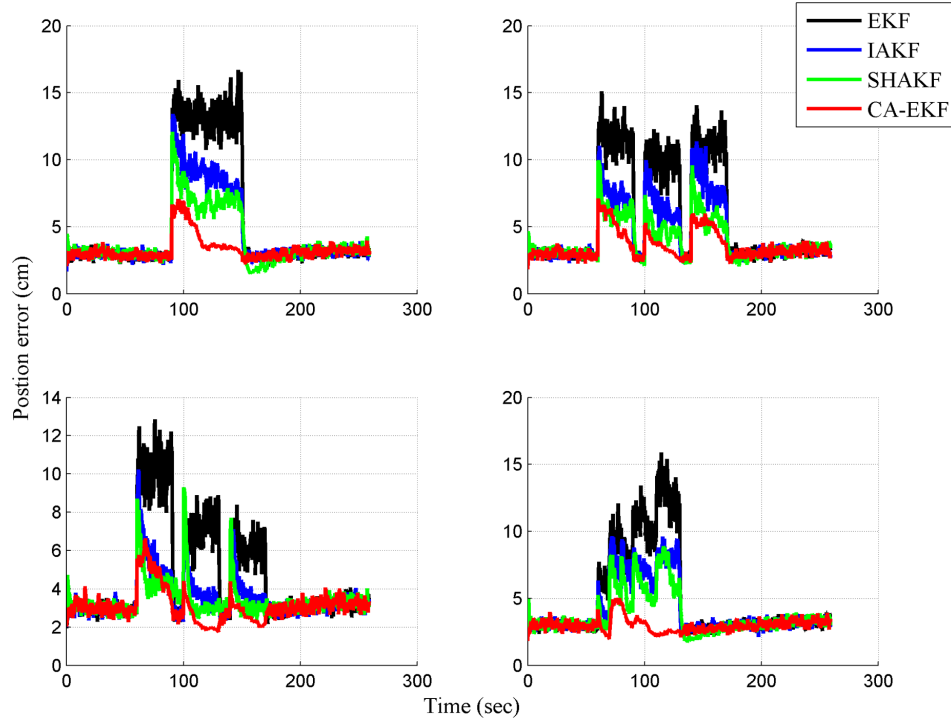


FIGURE 4 | Position error over time for different filtering techniques—Traj1. Top left: Simultaneous—three anchors. Top right: Simultaneous—two anchors. Bottom left: Isolated. Bottom right: Combined.

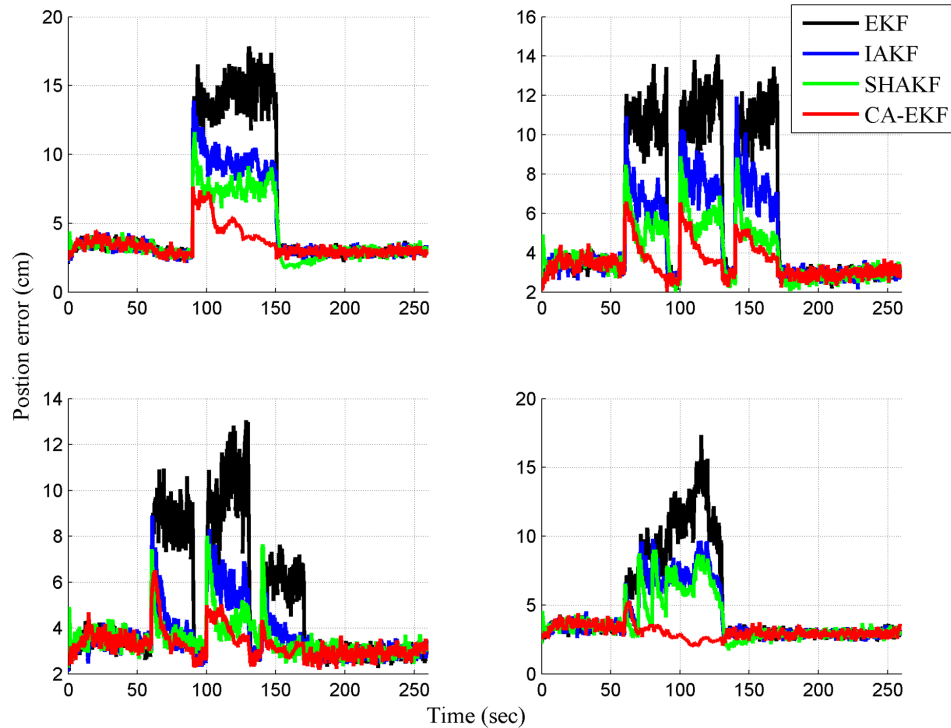


FIGURE 5 | Position error over time for different filtering techniques—Traj2. Top left: Simultaneous—three anchors. Top right: Simultaneous—two anchors. Bottom left: Isolated. Bottom right: Combined.

- *MSE and STD versus range error variance*: These metrics evaluate the impact of range measurement errors on positioning accuracy. MSE quantifies the average squared error, while STD measures the dispersion of errors, indicating the stability of each filtering technique under varying noise conditions.

4.4.1 | Position Error Results

Figures 4 and 5 compare the positioning error over time for different outlier scenarios using the proposed CA-EKF algorithm against the traditional EKF, IAKF, and SHAKF techniques.

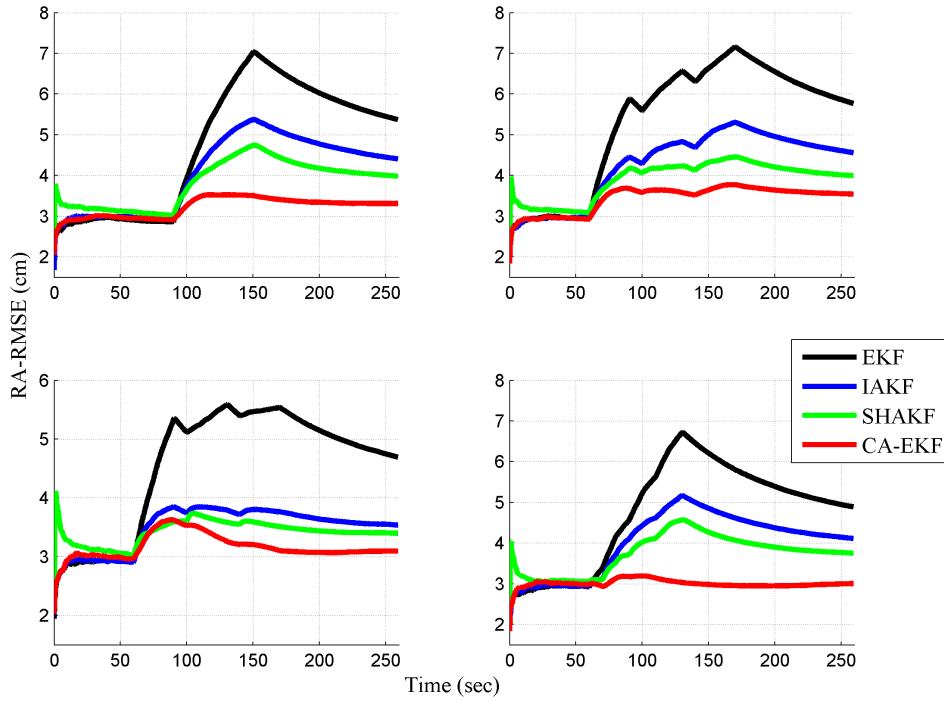


FIGURE 6 | RA-RMSE over time for different filtering techniques—Traj1. Top left: Simultaneous—three anchors. Top right: Simultaneous—two anchors. Bottom left: Isolated. Bottom right: Combined.

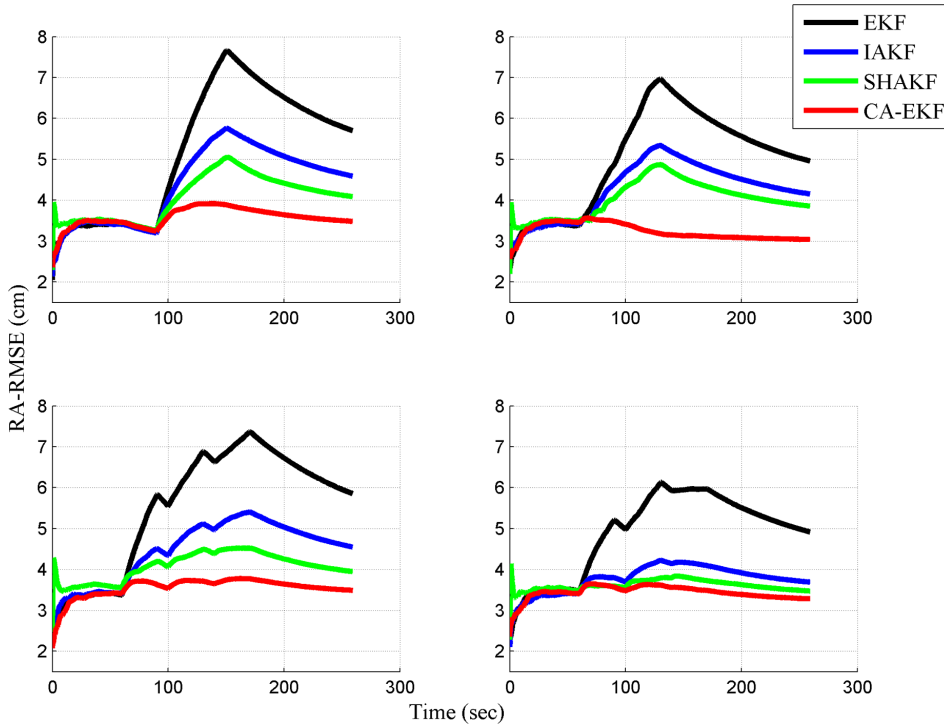


FIGURE 7 | RA-RMSE over time for different filtering techniques—Traj2. Top left: Simultaneous—three anchors. Top right: Simultaneous—two anchors. Bottom left: Isolated. Bottom right: Combined.

Monte Carlo simulations were conducted under the conditions in Table 1, considering two trajectory scenarios: Traj1 (smooth, nonmaneuvering motion) in Figure 4 and Traj2 (including turning maneuvers) in Figure 5. The four aforementioned outlier scenarios were examined, where outlier effects were modeled by amplifying the measurement noise over predefined time intervals, as described by Equation (26).

The results in Figure 4 clearly highlight the superiority of our proposed CA-EKF method over other filtering techniques in terms of accuracy, robustness, and error mitigation. Compared to EKF, IAKF, and SHAKF, CA-EKF significantly reduces positioning error, especially during critical time intervals (90–150s in the first scenario; 60–90, 100–130, and 140–170s in the second and third scenarios; and 60–129s in the last one), where conventional filters exhibit pronounced error peaks. Furthermore, the CA-EKF curve remains more stable and smoother, even in

challenging segments that cause sudden fluctuations in other filters. Notably, CA-EKF demonstrates exceptional robustness in scenarios with multiple simultaneous outliers, a condition under which conventional filters struggle to recover effectively. This resilience makes CA-EKF an optimal choice for high-precision tracking applications.

The results in Figure 5 further demonstrate the robustness of the CA-EKF algorithm, particularly in scenarios involving turning maneuvers. As observed previously, CA-EKF outperforms other filtering techniques by maintaining higher accuracy, significantly reducing positioning error, and ensuring better stability. Notably, during critical periods where conventional filters experience pronounced fluctuations, CA-EKF effectively limits error magnitude and quickly recovers reliable estimates. This performance is especially evident in time intervals where other methods struggle to adapt to perturbations. These findings highlight the effectiveness of CA-EKF for precise tracking in dynamic environments with varying motion patterns and significant disturbances.

TABLE 2 | Performance comparison in terms of RA-RMSE of filtering techniques.

Filter type	Max RA-RMSE (cm)	Last RA-RMSE (cm)
CA-EKF	~3.91	~3.54
SHAKF	~5.0	~4.0
IAKF	~5.8	~4.59
EKF	~7.66	~5.85

4.4.2 | RA-RMSE Results

Figures 6 and 7 present the RA-RMSE analysis for different filtering techniques across two trajectory scenarios. The running average of the RMSE over time provides a smoothed error evolution, allowing for a clearer comparison.

The results indicate that both figures exhibit approximately the same performance across all four scenarios, with CA-EKF

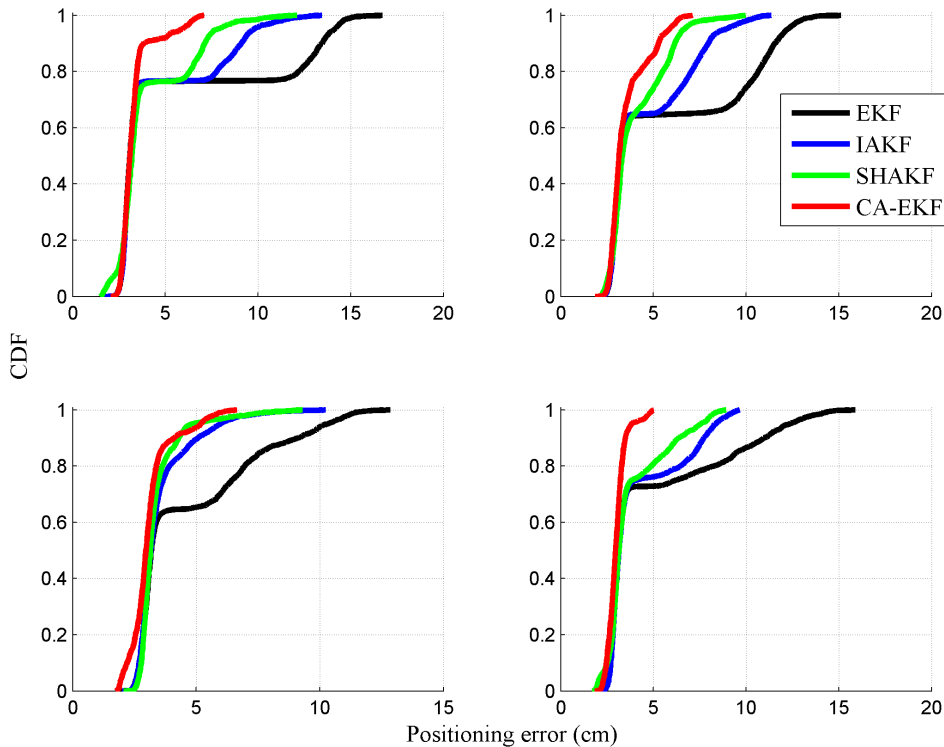


FIGURE 8 | CDF over positioning error for different filtering techniques—Traj1. Top left: Simultaneous—three anchors. Top right: Simultaneous—two anchors. Bottom left: Isolated. Bottom right: Combined.

consistently achieves lower RA-RMSE values compared to EKF, IAKF, and SHAKF. While conventional filters experience noticeable error accumulation over time, CA-EKF maintains a more stable and reduced error level, demonstrating its effectiveness in mitigating error propagation. This highlights its robustness and reliability for high-precision tracking applications.

As shown in Table 2, EKF exhibits the highest RA-RMSE, peaking at ~ 7.66 cm and stabilizing at ~ 5.85 cm. IAKF follows with a peak RA-RMSE of ~ 5.8 cm and a final error of ~ 4.59 cm, while SHAKF shows a slightly better performance with a peak of ~ 5.0 cm and a final value of ~ 4.0 cm. CA-EKF maintains the lowest error levels, with a peak RA-RMSE of ~ 3.91 cm and a final value of ~ 3.54 cm, making it the most suitable choice for high-precision tracking applications.

The consistent performance of CA-EKF across different scenarios highlights its robustness and reliability for tracking applications requiring high accuracy.

4.4.3 | CDF Results

Figures 8 and 9 illustrate the CDF analysis of positioning errors for various filtering techniques across different trajectory scenarios. These results provide a comparative assessment of each method's accuracy and reliability. The data for these figures were derived from positioning error evaluations conducted over multiple trajectory scenarios. By computing the CDF for each filtering technique, this analysis quantifies the proportion of errors falling below specific thresholds, offering valuable insights into the overall performance and effectiveness of each method.

The table below summarizes the key observations concerning this performance study metric.

As shown in Figures 8 and 9 and summarized in Table 3, the interpretation of the results reveals a clear ranking among the filtering techniques in terms of accuracy and reliability. CA-EKF demonstrates the best performance, with 99% of errors below 7.65 cm, making it the most precise and reliable method. Its positioning errors remain consistently low across all scenarios, ensuring superior accuracy. SHAKF follows closely, with 99% of errors under 12.09 cm. While it performs well, it is slightly less effective than CA-EKF in minimizing errors. IAKF exhibits moderate performance, with 99% of errors below 13.9 cm, showing slightly higher positioning errors compared to SHAKF. Finally, EKF is the weakest performer, with 99% of errors below 17.85 cm, demonstrating significant error variations and making it the least suitable for high-precision applications. In conclusion, the CDF analysis confirms CA-EKF as the most effective filtering technique, significantly outperforming the others in terms of accuracy and error reduction.

TABLE 3 | Performance comparison in terms of CDF of filtering techniques.

Filtering technique	% of error below threshold
CA-EKF	$\sim 99\%$ below 7.65 cm
SHAKF	$\sim 99\%$ below 12.09 cm
IAKF	$\sim 99\%$ below 13.9 cm
EKF	$\sim 99\%$ below 17.85 cm

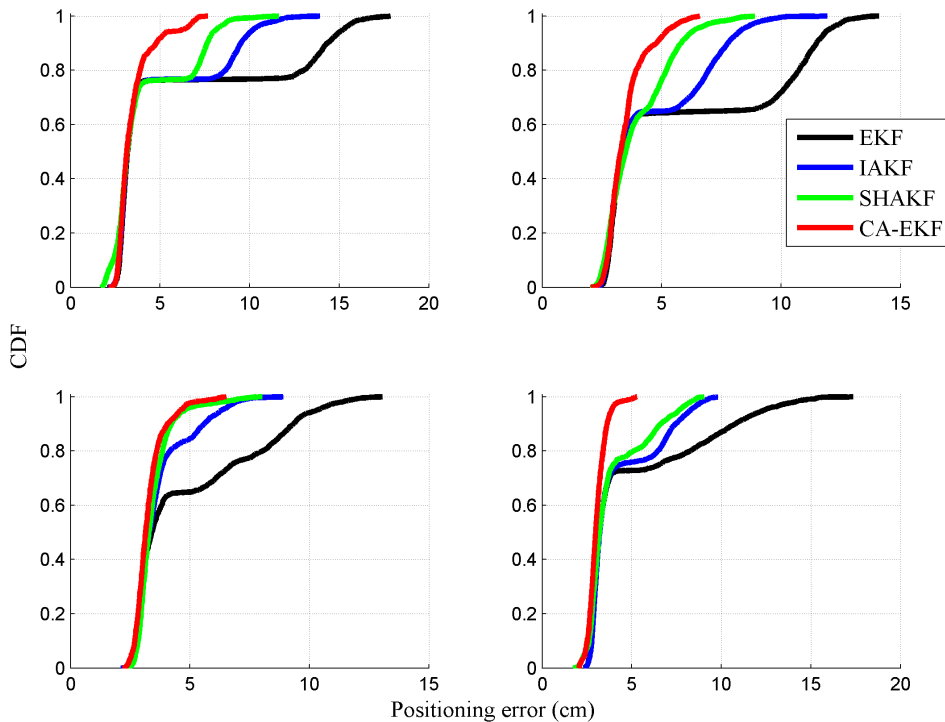


FIGURE 9 | CDF over positioning error for different filtering techniques—Traj2. Top left: Simultaneous—three anchors. Top right: Simultaneous—two anchors. Bottom left: Isolated. Bottom right: Combined.

4.4.4 | MSE Results

Figures 10 and 11 present the MSE analysis for two trajectory scenarios, illustrating how MSE evolves with increasing positioning error variance across different filtering techniques.

These results provide valuable insights into each method's robustness and accuracy. The curves were derived from multiple trajectory evaluations under varying noise conditions, enabling a comprehensive comparison of error propagation and stability.

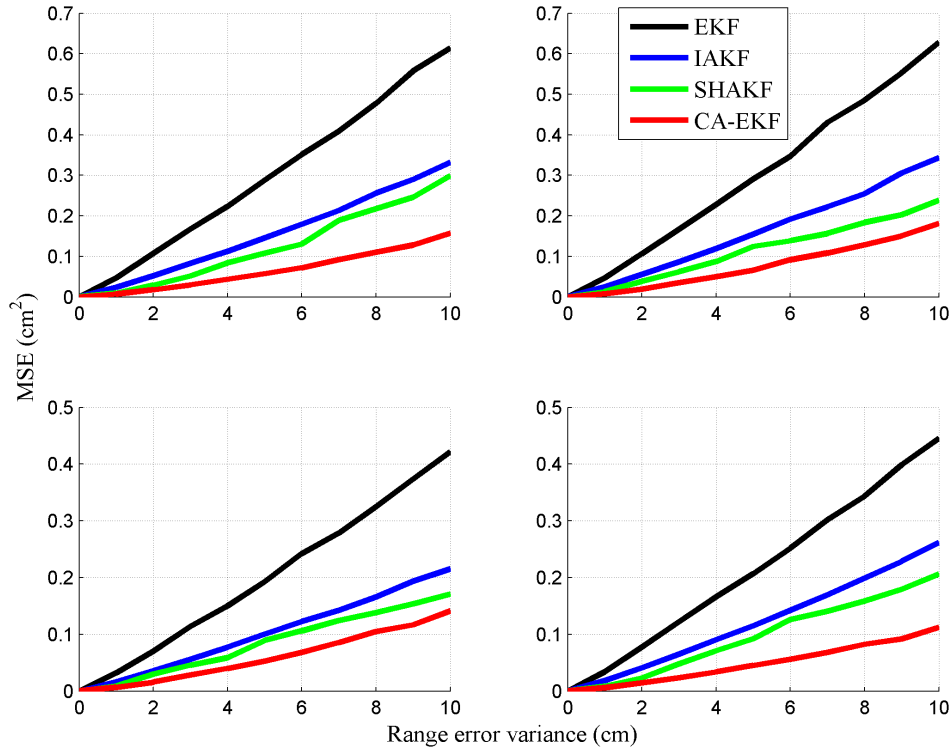


FIGURE 10 | MSE over range error variance for different filtering techniques—Traj1. Top left: Simultaneous—three anchors. Top right: Simultaneous—two anchors. Bottom left: Isolated. Bottom right: Combined.

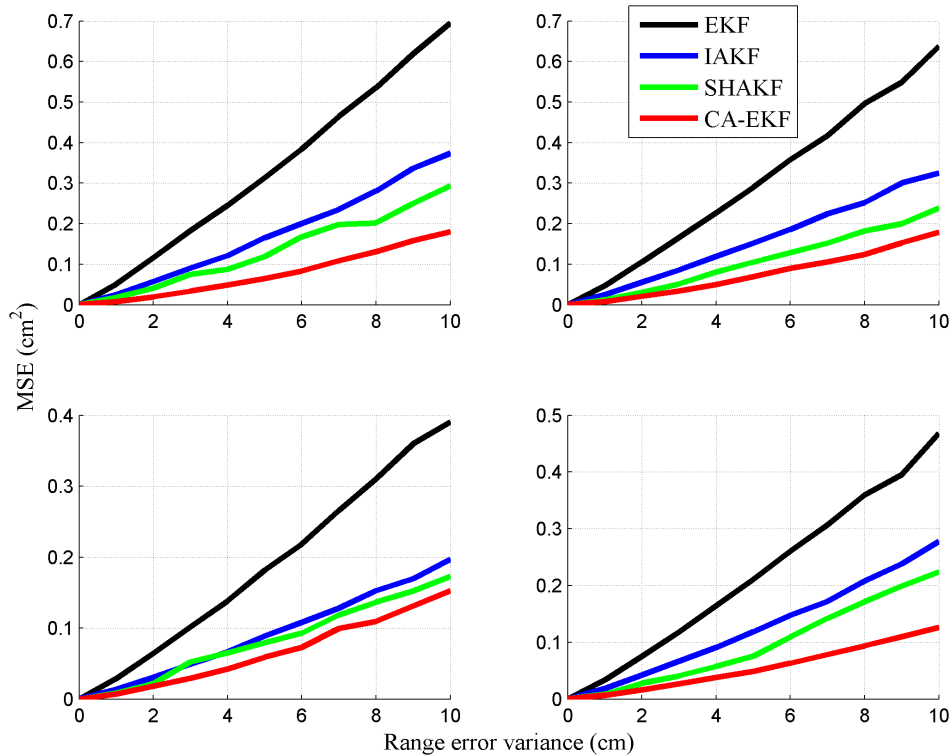


FIGURE 11 | MSE over range error variance for different filtering techniques—Traj2. Top left: Simultaneous—three anchors. Top right: Simultaneous—two anchors. Bottom left: Isolated. Bottom right: Combined.

As shown in these two figures, the MSE increases as the positioning error variance grows but at different rates depending on the filtering technique. The CA-EKF consistently achieves the best performance, maintaining the lowest MSE across all noise levels. This confirms its superior stability and accuracy compared to the other methods.

Table 4 provides a comparative analysis of the maximum MSE observed for each filtering technique, highlighting their respective performance with increasing positioning error variance.

As illustrated in this table, the CA-EKF achieves the lowest MSE ($\sim 0.18 \text{ cm}^2$), confirming its superior stability and accuracy in handling positioning errors. The SHAKF follows closely with a slightly higher MSE ($\sim 0.30 \text{ cm}^2$), demonstrating strong performance but not reaching the same level of precision as CA-EKF. The IAKF exhibits a more noticeable increase in MSE ($\sim 0.37 \text{ cm}^2$), indicating that it is more affected by positioning noise compared to CA-EKF and SHAKF. Finally, the EKF shows the highest MSE ($\sim 0.69 \text{ cm}^2$), making it the least reliable

TABLE 4 | Performance comparison in terms of MSE of filtering techniques.

Filtering technique	Max MSE (cm^2)
CA-EKF	~ 0.18
SHAKF	~ 0.30
IAKF	~ 0.37
EKF	~ 0.69

technique due to its pronounced sensitivity to increasing positioning errors. In conclusion, the CA-EKF filtering technique proves to be the most effective, offering the lowest MSE and the best stability under varying noise conditions.

4.4.5 | STD Results

Figures 12 and 13 show the evolution of STD as a function of range error variance for both targets across different filtering techniques. As with the MSE analysis, STD increases with higher positioning error variance, revealing distinct performance characteristics for each filter. As shown in Table 5, which presents the maximum STD values for each filtering technique, CA-EKF consistently maintains the lowest STD across all noise levels, demonstrating superior stability and accuracy.

These results confirm CA-EKF as the most stable and accurate method, achieving the lowest STD. SHAKF follows closely with slightly higher errors, while IAKF shows moderate performance but is more affected by noise. EKF, with the highest STD, demonstrates the least effectiveness due to its sensitivity to positioning errors.

4.4.6 | Convergence Analysis of CA-EKF

The convergence of CA-EKF is enhanced by its adaptive covariance update mechanism, which continuously refines the process and measurement noise parameters based on residual analysis. Unlike EKF, which relies on fixed noise models, and IAKF/SHAKF, which apply heuristic updates, CA-EKF leverages

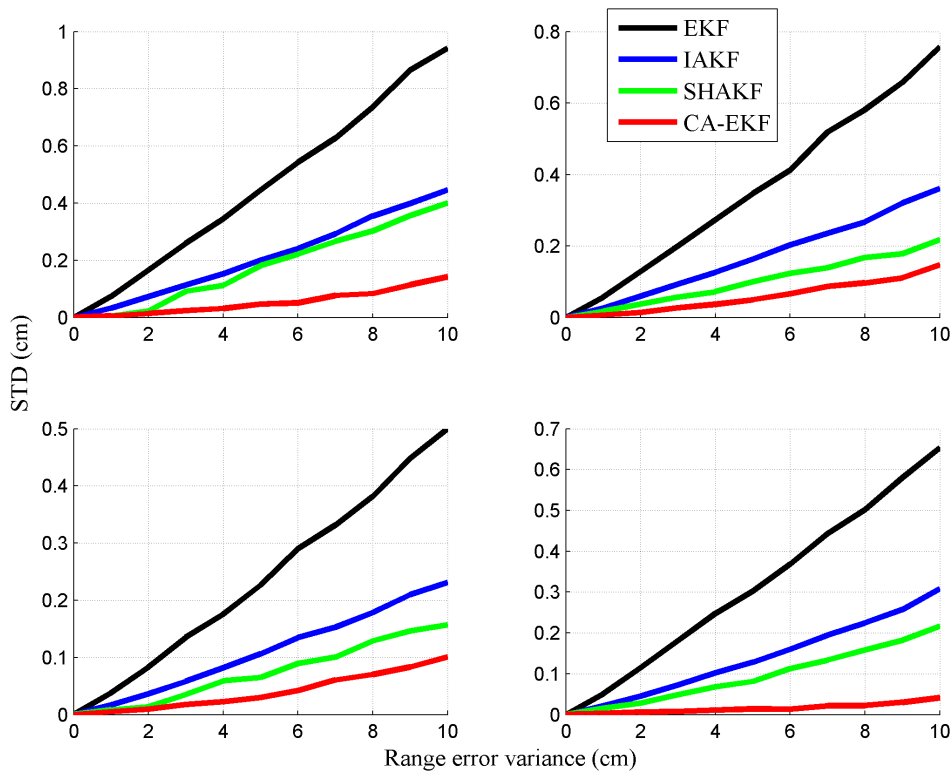


FIGURE 12 | STD over range error variance for different filtering techniques—Traj1. Top left: Simultaneous—three anchors. Top right: Simultaneous—two anchors. Bottom left: Isolated. Bottom right: Combined.

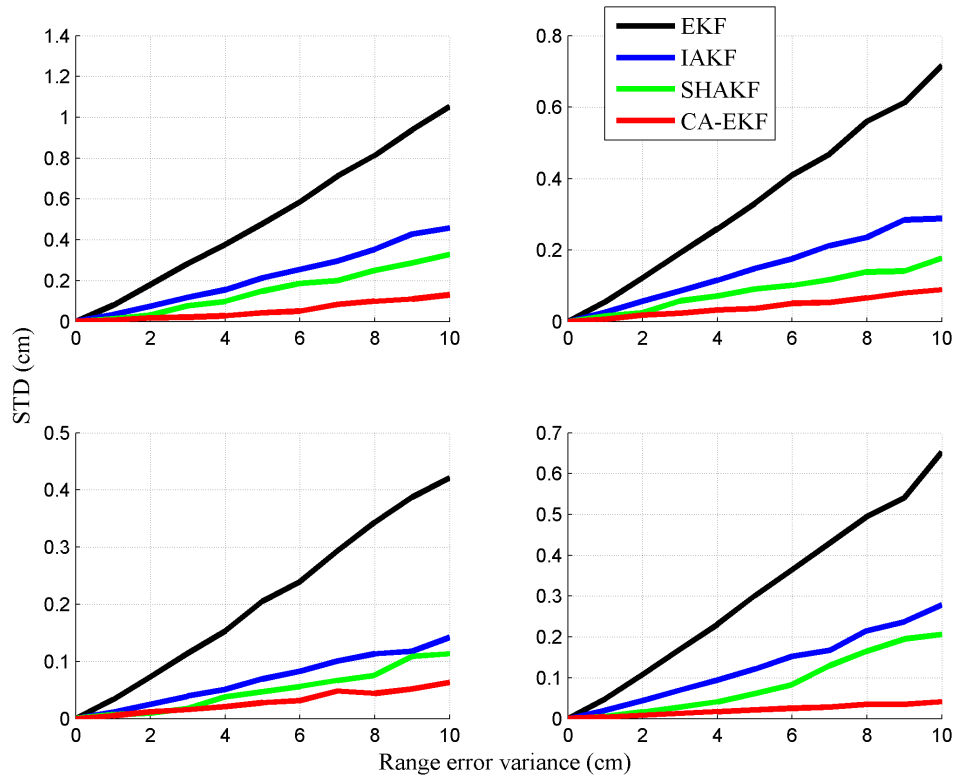


FIGURE 13 | STD over range error variance for different filtering techniques—Traj2. Top left: Simultaneous—three anchors. Top right: Simultaneous—two anchors. Bottom left: Isolated. Bottom right: Combined.

TABLE 5 | Performance comparison in terms of STD of filtering techniques.

Filtering technique	Max STD (cm)
CA-EKF	~0.14
SHAKF	~0.40
IAKF	~0.46
EKF	~1.05

cumulative residual statistics to dynamically adjust covariance values. This approach accelerates the stabilization of the estimated state and mitigates divergence risks in complex scenarios. As a result, CA-EKF achieves a faster convergence rate, particularly in the presence of outliers and dynamic motion, making it more suitable for real-time indoor localization applications.

5 | Conclusions

In this paper, we proposed an innovative CA-EKF technique to enhance the positioning accuracy and robustness of UWB localization systems in indoor environments. The proposed method addresses the limitations of traditional EKF approaches by adaptively updating the process and measurement noise covariance matrices using cumulative averages of residual and innovation vectors. This adaptation effectively mitigates the impact of outliers, prevents filter divergence, and ensures more stable state estimation.

Simulation results under various outlier scenarios and nonlinear motion trajectories demonstrated that CA-EKF achieves superior performance compared to traditional EKF, IAKF, and SHAKF techniques. Specifically, it significantly improves positioning accuracy by reducing the RA-RMSE, ensuring smoother error evolution, and minimizing deviations over time. The robustness of CA-EKF against both isolated and simultaneous outliers is particularly notable, especially in nonlinear trajectories, where it consistently outperforms alternative methods. Furthermore, the CDF analysis confirms that 99% of positioning errors remain below 7.65 cm, underscoring the effectiveness of CA-EKF in real-world indoor localization scenarios. These results validate CA-EKF as a reliable and adaptive solution for UWB-based positioning systems, offering enhanced precision and stability for dynamic tracking applications.

Despite these advantages, some practical considerations must be addressed. The efficiency of the adaptive noise tuning mechanism depends on the stability of residuals, which may require further refinements in highly dynamic environments with abrupt noise variations. Additionally, the iterative noise covariance update introduces a slight increase in computational complexity compared to conventional EKF approaches. While this overhead remains manageable for most modern embedded systems, future work will focus on optimizing matrix computations and exploring adaptive update strategies to ensure real-time feasibility.

Future work will focus on integrating machine learning techniques to enhance the adaptive behavior of the CA-EKF, especially in terms of dynamic noise modeling and anomaly

detection. Moreover, real-world testing in complex indoor environments will be carried out to validate the proposed approach and confirm its effectiveness in various positioning and tracking applications.

Data Availability Statement

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

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